IE 7374
Machine Learning in Engineering

Northeastern University, Fall 2021

Problem Set 1, Due: Oct 16, 2019

Problem Set Rules:

1. Each student should hand in an individual problem set at the beginning of class.
2. Discussing problem sets with other students is permitted. Copying from another person or solution set is *not* permitted.
3. Late assignments will *not* be accepted. No exceptions.

1. (Total: 15 points)

This exercise involves the Auto data set. Make sure that the missing values have been removed from the data.

(a) (2 points) Which of the predictors are quantitative, and which are qualitative?

Quantitative:
mpg cylinders displacement horsepower weight acceleration year

Qualitative:
origin name

(b) (2 points) What is the range (e.g., minimum and maximum) of each quantitative predictor?

```
df.describe()
```

	mpg	cylinders	displacement	horsepower	weight	acceleration	year	origin
count	392.000000	392.000000	392.000000	392.000000	392.000000	392.000000	392.000000	392.000000
mean	23.445918	5.471939	194.411990	104.469388	2977.584184	15.541327	75.979592	1.576531
std	7.805007	1.705783	104.644004	38.491160	849.402560	2.758864	3.683737	0.805518
min	9.000000	3.000000	68.000000	46.000000	1613.000000	8.000000	70.000000	1.000000
25%	17.000000	4.000000	105.000000	75.000000	2225.250000	13.775000	73.000000	1.000000
50%	22.750000	4.000000	151.000000	93.500000	2803.500000	15.500000	76.000000	1.000000
75%	29.000000	8.000000	275.750000	126.000000	3614.750000	17.025000	79.000000	2.000000
max	46.600000	8.000000	455.000000	230.000000	5140.000000	24.800000	82.000000	3.000000

(c) (2 points) What is the mean and standard deviation of each quantitative predictor?

```
df.describe()
```

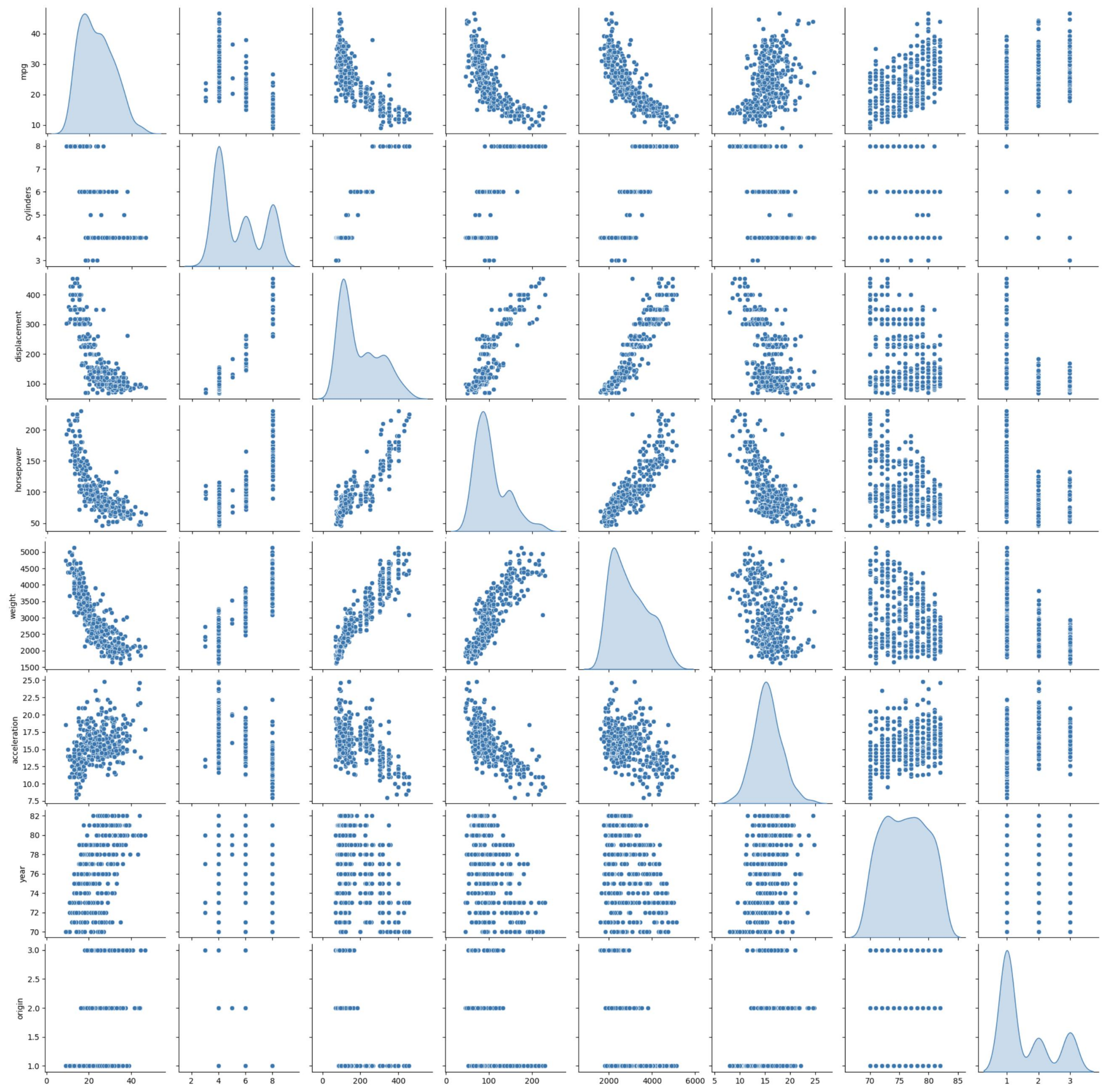
	mpg	cylinders	displacement	horsepower	weight	acceleration	year	origin
count	392.000000	392.000000	392.000000	392.000000	392.000000	392.000000	392.000000	392.000000
mean	23.445918	5.471939	194.411990	104.469388	2977.584184	15.541327	75.979592	1.576531
std	7.805007	1.705783	104.644004	38.491160	849.402560	2.758864	3.683737	0.805518
min	9.000000	3.000000	68.000000	46.000000	1613.000000	8.000000	70.000000	1.000000
25%	17.000000	4.000000	105.000000	75.000000	2225.250000	13.775000	73.000000	1.000000
50%	22.750000	4.000000	151.000000	93.500000	2803.500000	15.500000	76.000000	1.000000
75%	29.000000	8.000000	275.750000	126.000000	3614.750000	17.025000	79.000000	2.000000
max	46.600000	8.000000	455.000000	230.000000	5140.000000	24.800000	82.000000	3.000000

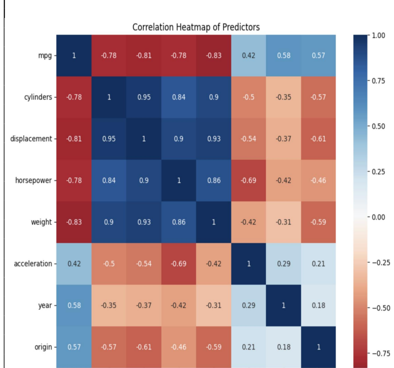
(d) (3 points) Now remove the 10th through 85th observations. What is the range, mean, and standard deviation of each predictor in the subset of the data that remains?

```
df_1_removed.describe()
```

	mpg	cylinders	displacement	horsepower	weight	acceleration	year	origin
count	317.000000	317.000000	317.000000	317.000000	317.000000	317.000000	317.000000	317.000000
mean	24.368454	5.381703	187.753943	100.955836	2939.643533	15.718297	77.132492	1.599369
std	7.880898	1.658135	99.939488	35.895567	812.649629	2.693813	3.110026	0.819308
min	11.000000	3.000000	68.000000	46.000000	1649.000000	8.500000	70.000000	1.000000
25%	18.000000	4.000000	101.000000	75.000000	2215.000000	14.000000	75.000000	1.000000
50%	23.900000	4.000000	146.000000	90.000000	2795.000000	15.500000	77.000000	1.000000
75%	30.500000	6.000000	250.000000	115.000000	3520.000000	17.300000	80.000000	2.000000
max	46.600000	8.000000	455.000000	230.000000	4997.000000	24.800000	82.000000	3.000000

(e) (3 points) Using the full data set, investigate the predictors graphically, using scatterplots or other tools of your choice. Create some plots highlighting the relationships among the predictors. Comment on your findings.





We can identify several pairs of predictors that are most likely correlated from the graph, including:
(mpg, cylinders), (mpg, displacement), (mpg, horsepower), (mpg, weight), (mpg, year), (mpg, origin), (cylinders, displacement), (cylinders, horsepower), (cylinders, weight), etc.

(f) (3 points) Suppose that we wish to predict gas mileage (mpg) on the basis of the other variables. Do your plots suggest that any of the other variables might be useful in predicting mpg? Justify your answer.

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                    mpg   R-squared:                       0.821
Model:                            OLS   Adj. R-squared:                  0.818
Method:                 Least Squares   F-statistic:                     252.4
Date:                Fri, 20 Feb 2026   Prob (F-statistic):          2.04e-139
Time:                        20:14:29   Log-Likelihood:                -1023.5
No. Observations:                 392   AIC:                             2063.
Df Residuals:                     384   BIC:                             2095.
Df Model:                           7
Covariance Type:            nonrobust
================================================================================
                   coef    std err          t      P>|t|      [0.025      0.975]
--------------------------------------------------------------------------------
const          -17.2184      4.644     -3.707      0.000     -26.350      -8.087
cylinders       -0.4934      0.323     -1.526      0.128      -1.129       0.142
displacement     0.0199      0.008      2.647      0.008       0.005       0.035
horsepower      -0.0170      0.014     -1.230      0.220      -0.044       0.010
weight          -0.0065      0.001     -9.929      0.000      -0.008      -0.005
acceleration     0.0806      0.099      0.815      0.415      -0.114       0.275
year             0.7508      0.051     14.729      0.000       0.651       0.851
origin           1.4261      0.278      5.127      0.000       0.879       1.973
==============================================================================
Omnibus:                       31.906   Durbin-Watson:                   1.309
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               53.100
Skew:                           0.529   Prob(JB):                     2.95e-12
Kurtosis:                       4.460   Cond. No.                     8.59e+04
==============================================================================
```

As the plots show, all other variables may be useful and contribute the prediction.
But after we try a Multiple Linear Regression, we can find p-value of each variables. The cylinders, horsepower, and acceleration has a huge p-value which means these don't contribute much.

2. (Total: 20 points) This exercise involves the Boston housing data set.

(a) (2 points) How many rows are in this data set? How many columns? What do the rows and columns represent?

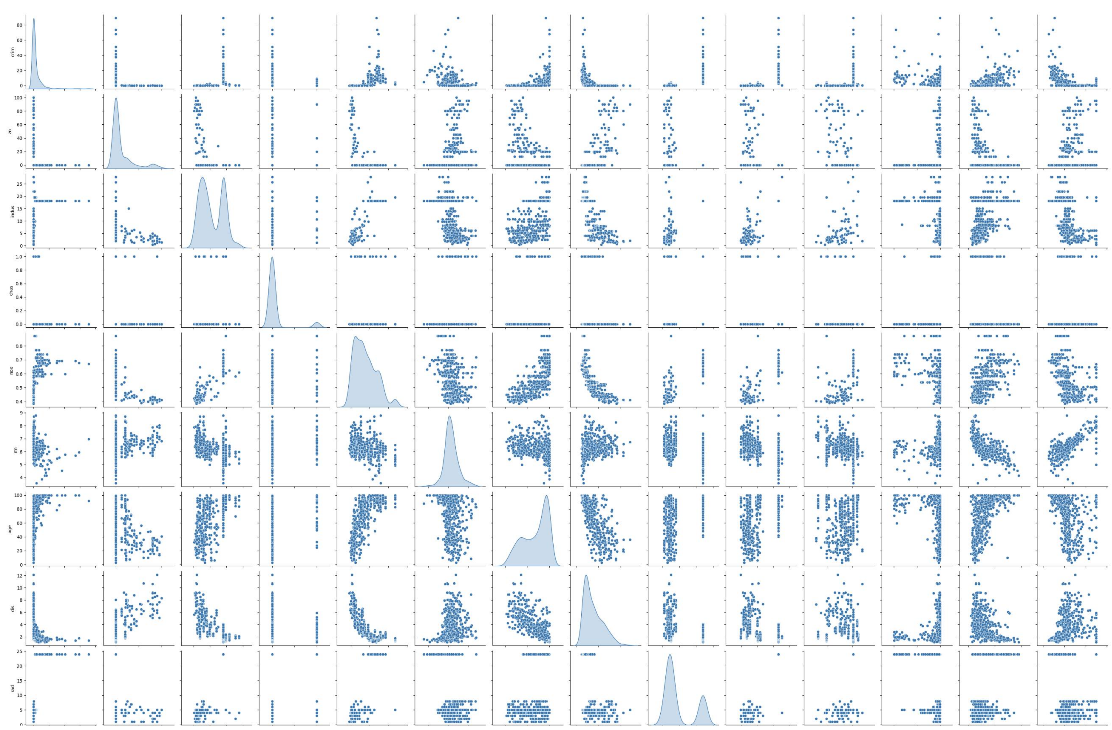
506 rows.
15 columns.

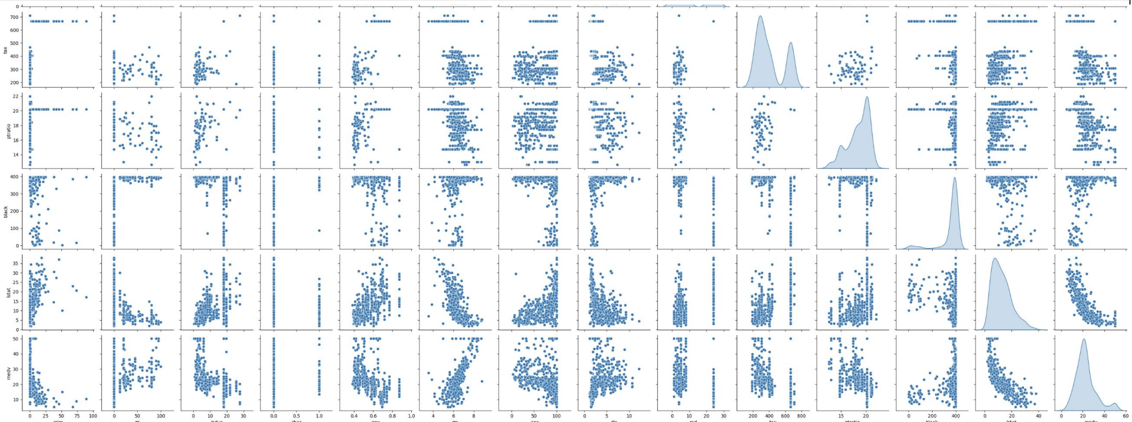
Each row means a data point.

First column is index, and other columns means the variables, including crim, zn, indus, etc.

```
RangeIndex: 506 entries, 0 to 505
Data columns (total 15 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Unnamed: 0  506 non-null    int64
 1   crim        506 non-null    float64
 2   zn          506 non-null    float64
 3   indus       506 non-null    float64
 4   chas        506 non-null    int64
 5   nox         506 non-null    float64
 6   rm          506 non-null    float64
 7   age         506 non-null    float64
 8   dis         506 non-null    float64
 9   rad         506 non-null    int64
 10  tax         506 non-null    int64
 11  ptratio     506 non-null    float64
 12  black       506 non-null    float64
 13  lstat       506 non-null    float64
 14  medv        506 non-null    float64
dtypes: float64(11), int64(4)
```

(b) (3 points) Make some pairwise scatterplots of the predictors (columns) in this data set. Describe your findings.





My findings are that there are serval pairs that most likely correlated, including: (crim, age), (crim, dis), (crim, medv), (zn, indus), etc.

And there are many pairs we can't tell if they are correlated or not without further calculation.

(c) (2 points) Are any of the predictors associated with per capita crime rate? If so, explain the relationship.

According to the plots, we can see that a low zn, high indus, 0 of chas, high nox, high age, low dis, high rad, high tax, high ptradio, low medv may lead to a high per cappita crime rate. And rm, black, lstat less likely have a impact to rate.

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                   crim   R-squared:                       0.454
Model:                            OLS   Adj. R-squared:                  0.440
Method:                 Least Squares   F-statistic:                     31.47
Date:                Fri, 20 Feb 2026   Prob (F-statistic):           1.57e-56
Time:                        21:31:28   Log-Likelihood:                -1653.3
No. Observations:                 506   AIC:                             3335.
Df Residuals:                     492   BIC:                             3394.
Df Model:                          13
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const         17.0332      7.235      2.354      0.019       2.818      31.248
zn             0.0449      0.019      2.394      0.017       0.008       0.082
indus         -0.0639      0.083     -0.766      0.444      -0.228       0.100
chas          -0.7491      1.180     -0.635      0.526      -3.068       1.570
nox          -10.3135      5.276     -1.955      0.051     -20.679       0.052
rm             0.4301      0.613      0.702      0.483      -0.774       1.634
age            0.0015      0.018      0.081      0.935      -0.034       0.037
dis           -0.9872      0.282     -3.503      0.001      -1.541      -0.433
rad            0.5882      0.088      6.680      0.000       0.415       0.761
tax           -0.0038      0.005     -0.733      0.464      -0.014       0.006
ptratio       -0.2711      0.186     -1.454      0.147      -0.637       0.095
black         -0.0075      0.004     -2.052      0.041      -0.015      -0.000
lstat          0.1262      0.076      1.667      0.096      -0.023       0.275
medv          -0.1989      0.061     -3.287      0.001      -0.318      -0.080
==============================================================================
Omnibus:                      666.613   Durbin-Watson:                   1.519
Prob(Omnibus):                  0.000   Jarque-Bera (JB):            84887.625
Skew:                           6.617   Prob(JB):                         0.00
Kurtosis:                      65.058   Cond. No.                     1.58e+04
==============================================================================
```

After the regression, we can find that:

zn, dis, rad, black, and medv are associated with per capita crime rate

zn and rad have a positive effect, while others are negative

(d) (3 points) Do any of the suburbs of Boston appear to have particularly high crime rates? Tax rates? Pupil-teacher ratios? Comment on the range of each predictor.

```
df_2[['crim', 'tax', 'ptratio']].describe()
```

	crim	tax	ptratio
count	506.000000	506.000000	506.000000
mean	3.613524	408.237154	18.455534
std	8.601545	168.537116	2.164946
min	0.006320	187.000000	12.600000
25%	0.082045	279.000000	17.400000
50%	0.256510	330.000000	19.050000
75%	3.677083	666.000000	20.200000
max	88.976200	711.000000	22.000000

There are some suburbs that have particularly high crime rates. We can see that we have mean value of 3.6, and 75% of suburbs have crime rates lower than 3.7, but we can find a max value of 88.98 which is far beyond normal.

The tax and ptratio is good and they have no particularly high values.

(e) (2 points) How many of the suburbs in this data set bound the Charles river?

```
df_2[df_2['chas'] == 1].count()
```

...

	0
Unnamed: 0	35
crim	35

35

(f) (2 points) What is the median pupil-teacher ratio among the towns in this data set?

```
df_2['ptratio'].median()
```

```
... 19.05
```

19.05

(g) (3 points) Which suburb of Boston has lowest median value of owner- occupied homes? What are the values of the other predictors for that suburb, and how do those values compare to the overall ranges for those predictors? Comment on your findings.

```
df_2[df_2['medv'] == df_2['medv'].min()]
```

	Unnamed: 0	crim	zn	indus	chas	nox	rm	age	dis	rad	tax	ptratio	black	lstat	medv
398	399	38.3518	0.0	18.1	0	0.693	5.453	100.0	1.4896	24	666	20.2	396.90	30.59	5.0
405	406	67.9208	0.0	18.1	0	0.693	5.683	100.0	1.4254	24	666	20.2	384.97	22.98	5.0

These two have lowest value.

They both have a extremely high crime rates, high nox(> 75%), low rm(< 25%), max age, low dis(< 25%), max rad, high tax(> 75%), high ptratio (> 75%), high lstat(> 75%)

(h) (3 points) In this data set, how many of the suburbs average more than seven rooms per dwelling? More than eight rooms per dwelling? Comment on the suburbs that average more than eight rooms per dwelling.

```
df_2[df_2['rm'] > 7].shape[0]
```

```
64
```

>7: 64

```
df_2[df_2['rm'] > 8].shape[0]
```

```
13
```

>8: 13

```
df_2[df_2['rm'] > 8].drop(columns='Unnamed: 0').describe()
```

	crim	zn	indus	chas	nox	rm	age	dis	rad	tax	ptratio	black	lstat	medv
count	13.000000	13.000000	13.000000	13.000000	13.000000	13.000000	13.000000	13.000000	13.000000	13.000000	13.000000	13.000000	13.000000	13.000000
mean	0.718795	13.615385	7.078462	0.153846	0.539238	8.348538	71.538462	3.430192	7.461538	325.076923	16.361538	385.210769	4.310000	44.200000
std	0.901640	26.298094	5.392767	0.375534	0.092352	0.251261	24.608723	1.883955	5.332532	110.971063	2.410580	10.529359	1.373566	8.092383
min	0.020090	0.000000	2.680000	0.000000	0.416100	8.034000	8.400000	1.801000	2.000000	224.000000	13.000000	354.550000	2.470000	21.900000
25%	0.331470	0.000000	3.970000	0.000000	0.504000	8.247000	70.400000	2.288500	5.000000	264.000000	14.700000	384.540000	3.320000	41.700000
50%	0.520140	0.000000	6.200000	0.000000	0.507000	8.297000	78.300000	2.894400	7.000000	307.000000	17.400000	386.860000	4.140000	48.300000
75%	0.578340	20.000000	6.200000	0.000000	0.605000	8.398000	86.500000	3.651900	8.000000	307.000000	17.400000	389.700000	5.120000	50.000000
max	3.474280	95.000000	19.580000	1.000000	0.718000	8.780000	93.900000	8.906700	24.000000	666.000000	20.200000	396.900000	7.440000	50.000000

They have lower crime rates, higher lstat and medv. Even the highest crime rates is lower than 75% of all suburbs.

3. (Total: 24 points)

In this question, you should use the Carseats data set to predict the sales in a new store with Price=\$120, Advertising=\$10000, ShelveLoc = Good, 'Urban=Yes, US=Yes.

(a) (3 points) Fit a multiple regression model to predict Sales using Price, Advertising Urban, and US. Write out the model in equation form, being careful to handle the qualitative variables properly.

```
...                            OLS Regression Results
    ==============================================================================
    Dep. Variable:                  Sales   R-squared:                       0.282
    Model:                            OLS   Adj. R-squared:                  0.275
    Method:                 Least Squares   F-statistic:                     38.77
    Date:                Fri, 20 Feb 2026   Prob (F-statistic):           2.21e-27
    Time:                        22:44:10   Log-Likelihood:                -916.11
    No. Observations:                 400   AIC:                             1842.
    Df Residuals:                     395   BIC:                             1862.
    Df Model:                           4
    Covariance Type:            nonrobust
    ===============================================================================
                      coef    std err          t      P>|t|      [0.025      0.975]
    -------------------------------------------------------------------------------
    const          13.0113      0.633     20.544      0.000      11.766      14.256
    Price          -0.0546      0.005    -10.710      0.000      -0.065      -0.045
    Advertising     0.1203      0.025      4.845      0.000       0.072       0.169
    Urban_num      -0.0388      0.264     -0.147      0.883      -0.558       0.481
    US_num          0.0585      0.345      0.170      0.865      -0.620       0.737
    ==============================================================================
    Omnibus:                        1.094   Durbin-Watson:                   1.964
    Prob(Omnibus):                  0.579   Jarque-Bera (JB):                0.983
    Skew:                           0.120   Prob(JB):                        0.612
    Kurtosis:                       3.036   Cond. No.                         629.
    ==============================================================================
```

Use 1 for "Yes", and 0 for "No"

y = 13.0113 - 0.0546*Price + 0.1203*Advertising - 0.0388*Urban + 0.0585*US

(b) (3 points) Provide an interpretation of each coefficient in the model. Be careful–some of the variables in the model are qualitative!

A lower price, a higher advertising, Urban of "No" and US of "Yes" can lead to a higher sales in the model.

(c) (4 points) Using the model from (a), predict sales in the new store and calculate 68% and 95% confidence intervals.

```
# Create a new DataFrame for the store's data
new_store_data = pd.DataFrame({
    'const': [1],  # Constant term for the intercept
    'Price': [120],
    'Advertising': [10], # 10k
    'Urban_num': [1],  # Urban=Yes
    'US_num': [1]      # US=Yes
})

# Predict the sales
predicted_sales = model_3.predict(new_store_data)[0]

print(f"Predicted Sales: {predicted_sales:.2f}")

# standerd deviation
rmse = np.sqrt(model_3.mse_resid)
print(f"RMSE: {rmse:f}")
print(f"68%: ({(predicted_sales - rmse):.2f}, {(predicted_sales + rmse):.2f})")
print(f"95%: ({(predicted_sales - rmse * 2):.2f}, {(predicted_sales + rmse * 2):.2f})")
```

```
Predicted Sales: 7.68
RMSE: 2.405189
68%: (5.28, 10.09)
95%: (2.87, 12.49)
```

(d) (3 points) Using the model from (a), what is the probability that sales will be greater than 12000 units in the new store?

```
[101]
from scipy.stats import norm

[103]
prob_greater_12 = norm.sf(12, loc=predicted_sales, scale=rmse)
print(f"Probability that Sales > 12,000: {prob_greater_12:.2%}")

Probability that Sales > 12,000: 3.63%
```

(e) (3 points) Using the model from (a), what is the probability that sales will be between 6000 and 10000 units in the new store?

```
[104]
prob_below_10 = norm.cdf(10, loc=predicted_sales, scale=rmse)
prob_below_6 = norm.cdf(6, loc=predicted_sales, scale=rmse)
prob_between = prob_below_10 - prob_below_6
print(f"Probability of Sales between 6,000 and 10,000: {prob_between:.2%}")

... Probability of Sales between 6,000 and 10,000: 59.02%
```

(f) (2 points) For which of the predictors can you reject the null hypothesis $H_0 : \beta_j = 0$?

	coef	std err	t	P>|t|	[0.0
const	13.0113	0.633	20.544	0.000	11.7
Price	-0.0546	0.005	-10.710	0.000	-0.0
Advertising	0.1203	0.025	4.845	0.000	0.0
Urban_num	-0.0388	0.264	-0.147	0.883	-0.5
US_num	0.0585	0.345	0.170	0.865	-0.6

Omnibus: 1.094 Durbin-Watson:

Price and Advertising

(g) (4 points) On the basis of your response to the previous question, fit a smaller model that only uses the predictors for which there is evidence of association with the outcome. Using this model, predict sales in the new store and calculate 68% and 95% confidence intervals.

```python
# Define the dependent variable
y_3 = df_3['Sales']

# Define the independent variables
X_3_cols = ['Price', 'Advertising']
X_3 = df_3[X_3_cols]

# Add a constant to the independent variables (required for Statsmodels OLS)
X_3 = sm.add_constant(X_3)

# Fit the multiple regression model
model_3_pa = sm.OLS(y_3, X_3).fit()

# Print the model summary
print(model_3_pa.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                  Sales   R-squared:                       0.282
Model:                            OLS   Adj. R-squared:                  0.278
Method:                 Least Squares   F-statistic:                     77.91
Date:                Fri, 20 Feb 2026   Prob (F-statistic):           2.87e-29
Time:                        23:30:22   Log-Likelihood:                -916.14
No. Observations:                 400   AIC:                             1838.
Df Residuals:                     397   BIC:                             1850.
Df Model:                           2
Covariance Type:            nonrobust
===============================================================================
                  coef    std err          t      P>|t|      [0.025      0.975]
-------------------------------------------------------------------------------
const          13.0034      0.607     21.428      0.000      11.810      14.196
Price          -0.0546      0.005    -10.755      0.000      -0.065      -0.045
Advertising     0.1231      0.018      6.809      0.000       0.088       0.159
==============================================================================
Omnibus:                        1.120   Durbin-Watson:                   1.964
Prob(Omnibus):                  0.571   Jarque-Bera (JB):                1.006
Skew:                           0.121   Prob(JB):                        0.605
Kurtosis:                       3.037   Cond. No.                         599.
==============================================================================
```

```python
# Create a new DataFrame for the store's data
new_store_data = pd.DataFrame({
    'const': [1],  # Constant term for the intercept
    'Price': [120],
    'Advertising': [10], # 10k
})

# Predict the sales
predicted_sales = model_3_pa.predict(new_store_data)[0]

print(f"Predicted Sales: {predicted_sales:.2f}")

# standerd deviation
rmse = np.sqrt(model_3_pa.mse_resid)
print(f"RMSE: {rmse:f}")
print(f"68%: ({(predicted_sales - rmse):.2f}, {(predicted_sales + rmse):.2f})")
print(f"95%: ({(predicted_sales - rmse * 2):.2f}, {(predicted_sales + rmse * 2):.2f})")
```

```
Predicted Sales: 7.68
RMSE: 2.399272
68%: (5.28, 10.08)
95%: (2.88, 12.48)
```

(h) (2 points) How well do the models in (a) and (g) fit the data?

Both models have a R^2 of 0.282.
It is not a high value so they don't fit the data very well.

The model in (g) has a higher Adj. R^2.
So we can say it works better than the model in (a).

4. (Total: 27 points) This problem involves the sales data set for Toyota Corolla, which can be found in the file ToyotaCorolla.csv. The data set contains 1436 observations on the following 10 variables.

   Price (in Dollars)

   Age (in months)

   Mileage

   FuelType Fuel Type (diesel, petrol, CNG)

   MetColor Metallic color (1=yes, 0=no)

   Automatic Automatic transmission (1=yes, 0=no)

   Displacement Engine displacement (in cu. inches)

   Doors Number of doors

   Weight (in pounds)

   Horsepower Engine horsepower

   (a) (2 points) Which of the predictors are quantitative, and which are qualitative?

   Quantitative
   Price, Age, Mileage, Displacement, Doors, Weight, Horsepower

   Qualitative:
   FuelType, MetColor, Automatic

   (b) (2 points) What is the range (i.e., min and max) of each quantitative predictor?

	Price	Age	Mileage	Horsepower	MetColor	Automatic	Displacement	Doors	Weight
count	1436.000000	1436.000000	1436.000000	1436.000000	1436.000000	1436.000000	1436.000000	1436.000000	1436.000000
mean	13199.266017	55.947075	42584.589136	101.502089	0.674791	0.055710	95.720752	4.033426	2364.442897
std	4461.162081	18.599988	23305.402480	14.981080	0.468616	0.229441	11.587120	0.952677	115.945453
min	5351.000000	1.000000	1.000000	69.000000	0.000000	0.000000	79.000000	2.000000	2205.000000
25%	10394.000000	44.000000	26719.000000	90.000000	0.000000	0.000000	85.000000	3.000000	2293.000000
50%	12177.000000	61.000000	39388.500000	110.000000	1.000000	0.000000	98.000000	4.000000	2359.000000
75%	14699.000000	70.000000	54072.000000	110.000000	1.000000	0.000000	98.000000	5.000000	2392.000000
max	39975.000000	80.000000	150993.000000	192.000000	1.000000	1.000000	122.000000	5.000000	3560.000000

(c) (2 points) What is the mean and standard deviation of each quantitative predictor?

```
df_4.describe()
```

	Price	Age	Mileage	Horsepower	MetColor	Automatic	Displacement	Doors	Weight
count	1436.000000	1436.000000	1436.000000	1436.000000	1436.000000	1436.000000	1436.000000	1436.000000	1436.000000
mean	13199.266017	55.947075	42584.589136	101.502089	0.674791	0.055710	95.720752	4.033426	2364.442897
std	4461.162081	18.599988	23305.402480	14.981080	0.468616	0.229441	11.587120	0.952677	115.945453
min	5351.000000	1.000000	1.000000	69.000000	0.000000	0.000000	79.000000	2.000000	2205.000000
25%	10394.000000	44.000000	26719.000000	90.000000	0.000000	0.000000	85.000000	3.000000	2293.000000
50%	12177.000000	61.000000	39388.500000	110.000000	1.000000	0.000000	98.000000	4.000000	2359.000000
75%	14699.000000	70.000000	54072.000000	110.000000	1.000000	0.000000	98.000000	5.000000	2392.000000
max	39975.000000	80.000000	150993.000000	192.000000	1.000000	1.000000	122.000000	5.000000	3560.000000

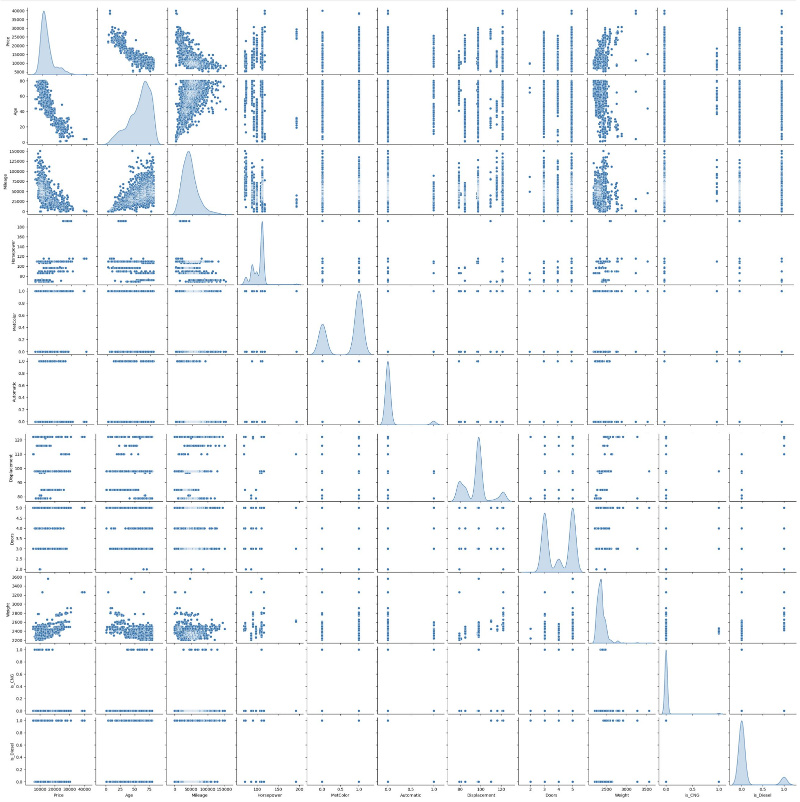
(d) (4 points) Investigate the predictors graphically, using scatterplots or other tools of your choice. Create some plots highlighting the relationships among the predictors. Comment on your findings.

```
df_4['FuelType'].value_counts()
```

	count
FuelType	
Petrol	1264
Diesel	155
CNG	17

dtype: int64

```
# Use the is_CNG and is_Diesel parameters, assigning values of 1 and 0 respectively, to indicate the FuelType.
df_4['is_CNG'] = df_4['FuelType'].apply(lambda x: 1 if x == 'CNG' else 0)
df_4['is_Diesel'] = df_4['FuelType'].apply(lambda x: 1 if x == 'Diesel' else 0)
df_4.head(10)
```



I find that there may be relationship in pairs such as:
(price, age), (price, Mileage), (price, Horsepower), etc.

(e) (4 points) Suppose that we wish to predict gas mileage (mpg) on the basis of the other variables. Do your plots suggest that any of the other variables might be useful in predicting mpg? Justify your answer.

According to the plots, I think these might be useful:
Price, Age

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                Mileage   R-squared:                       0.550
Model:                            OLS   Adj. R-squared:                  0.547
Method:                 Least Squares   F-statistic:                     174.5
Date:                Sat, 21 Feb 2026   Prob (F-statistic):          3.90e-239
Time:                        00:38:01   Log-Likelihood:                -15904.
No. Observations:                1436   AIC:                         3.183e+04
Df Residuals:                    1425   BIC:                         3.189e+04
Df Model:                          10
Covariance Type:            nonrobust
================================================================================
                   coef    std err          t      P>|t|      [0.025      0.975]
--------------------------------------------------------------------------------
const         1.519e+04   1.44e+04      1.053      0.293   -1.31e+04    4.35e+04
Price           -3.0120      0.244    -12.352      0.000      -3.490      -2.534
Age            115.8362     49.334      2.348      0.019      19.061     212.611
Horsepower      72.2040     70.531      1.024      0.306     -66.152     210.560
MetColor      -935.4768    892.954     -1.048      0.295   -2687.122     816.169
Automatic    -3440.4401   1873.009     -1.837      0.066   -7114.592     233.712
Displacement   114.5473    103.247      1.109      0.267     -87.986     317.080
Doors          967.0578    476.868      2.028      0.043      31.620    1902.496
Weight          15.1693      7.099      2.137      0.033       1.243      29.096
is_CNG        2.748e+04   3909.787      7.028      0.000    1.98e+04    3.51e+04
is_Diesel     2.878e+04   5071.651      5.675      0.000    1.88e+04    3.87e+04
==============================================================================
Omnibus:                       86.506   Durbin-Watson:                   0.680
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              169.942
Skew:                           0.411   Prob(JB):                     1.25e-37
Kurtosis:                       4.472   Cond. No.                     5.03e+05
==============================================================================
```

Then we can find that the p-value of Price, Age, Automatic, Doors, Weight, is_CNG, is_Diesel is smaller than 0.05, which means these variables are useful.

(f) (4 points) Fit a simple linear regression with Price as the response and Age as the predictor.

(i) Is there a relationship between the predictor and the response?

```python
y_4 = df_4['Price']
X_4 = df_4['Age']

# Add a constant for the intercept
X_4 = sm.add_constant(X_4)

# Fit the OLS model
model_4 = sm.OLS(y_4, X_4).fit()

# Display the summary to
print(model_4.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                  Price   R-squared:                       0.768
Model:                            OLS   Adj. R-squared:                  0.768
Method:                 Least Squares   F-statistic:                     4758.
Date:                Sat, 21 Feb 2026   Prob (F-statistic):               0.00
Time:                        00:42:07   Log-Likelihood:                -13054.
No. Observations:                1436   AIC:                         2.611e+04
Df Residuals:                    1434   BIC:                         2.612e+04
Df Model:                           1
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const       2.496e+04    179.699    138.910      0.000    2.46e+04    2.53e+04
Age         -210.2481      3.048    -68.978      0.000    -216.227    -204.269
==============================================================================
Omnibus:                      359.248   Durbin-Watson:                   1.214
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             2773.781
Skew:                           0.946   Prob(JB):                         0.00
Kurtosis:                       9.540   Cond. No.                         187.
==============================================================================
```

Yes.

(ii) How strong is the relationship between the predictor and the response?

We have a R^2 of 0.768, which means the relationship is quite strong.

(iii) What is the predicted price associated for a car with an age of 48 months? What are the associated 95% confidence intervals?

```
new_data = pd.DataFrame({
    'const': [1],  # Constant term for the intercept
    'Age': [48],
})

# Predict
predicted_res = model_4.predict(new_data)[0]

print(f"Predicted Result: {predicted_res:.2f}")

# standerd deviation
rmse = np.sqrt(model_4.mse_resid)
print(f"RMSE: {rmse:f}")
print(f"95%: ({(predicted_res - rmse * 2):.2f}, {(predicted_res + rmse * 2):.2f})")
```

```
Predicted Result: 14870.12
RMSE: 2147.624740
95%: (10574.87, 19165.37)
```

(g) (5 points) Fit a multiple linear regression with Price as the response and all other variables the predictors.

(i) Is there a relationship between the predictors and the response?

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                  Price   R-squared:                       0.869
Model:                            OLS   Adj. R-squared:                  0.868
Method:                 Least Squares   F-statistic:                     947.1
Date:                Sat, 21 Feb 2026   Prob (F-statistic):               0.00
Time:                        00:50:53   Log-Likelihood:                -12643.
No. Observations:                1436   AIC:                         2.531e+04
Df Residuals:                    1425   BIC:                         2.537e+04
Df Model:                          10
Covariance Type:            nonrobust
================================================================================
                   coef    std err          t      P>|t|      [0.025      0.975]
--------------------------------------------------------------------------------
const        -3671.9544   1487.015     -2.469      0.014   -6588.928    -754.981
Age           -149.9835      3.203    -46.823      0.000    -156.267    -143.700
Mileage         -0.0321      0.003    -12.352      0.000      -0.037      -0.027
Horsepower      73.7535      7.018     10.509      0.000      59.987      87.520
MetColor        70.2095     92.214      0.761      0.447    -110.681     251.100
Automatic      408.7614    193.314      2.114      0.035      29.551     787.972
Displacement   -79.1602     10.457     -7.570      0.000     -99.672     -58.648
Doors           -9.0320     49.307     -0.183      0.855    -105.754      87.690
Weight          11.1744      0.672     16.632      0.000       9.856      12.492
is_CNG       -1377.5453    408.997     -3.368      0.001   -2179.846    -575.244
is_Diesel     2615.5493    524.980      4.982      0.000    1585.732    3645.366
==============================================================================
Omnibus:                      273.419   Durbin-Watson:                   1.619
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             2803.262
Skew:                          -0.569   Prob(JB):                         0.00
Kurtosis:                       9.750   Cond. No.                     1.73e+06
==============================================================================
```

Yes. Prob (F-statistic) < 0.05

(ii) How strong is the relationship between the predictors and the response?

R^2 is 0.869. It is very strong.

(iii) Which predictors appear to have a statistically significant relationship to the response?

All except for the MetColor and Doors.

MetColor and Doors have a p-value > 0.05.

(iv) What does the coefficient for the age variable suggest? How accurate can you estimate the effect of age on price?

-149.9835

95% in range [-156.267, -143.700]

(v) What is the predicted price associated for a car with a mileage of 45000 miles, 48 months, diesel, automatic transmission, 4 doors, 2568 pounds, a displacement of 122 cu. inches, a horsepower of 90, and non-metallic color? What are the associated 95% confidence intervals?

```python
new_data = pd.DataFrame({
    'const': [1],  # Constant term for the intercept
    'Age': [48],
    'Mileage': [45000],
    'Horsepower': [90],
    'MetColor': [0],
    'Automatic': [1],
    'Displacement': [122],
    'Doors': [4],
    'Weight': [2568],
    'is_CNG': [0],
    'is_Diesel': [1]
})

# Predict
predicted_res = model_4.predict(new_data)[0]

print(f"Predicted Result: {predicted_res:.2f}")

# standerd deviation
rmse = np.sqrt(model_4.mse_resid)
print(f"RMSE: {rmse:f}")
print(f"95%: ({(predicted_res - rmse * 2):.2f}, {(predicted_res + rmse * 2):.2f})")
```

```
Predicted Result: 16348.12
RMSE: 1618.986411
95%: (13110.15, 19586.09)
```

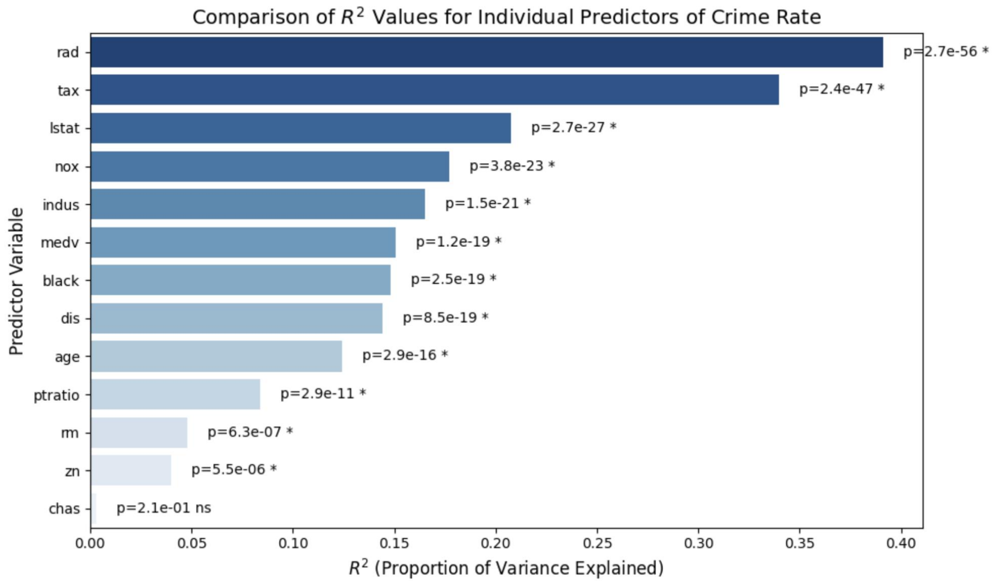
(h) (4 points) Which predictors predictors matter most for predicting the price for a car? (Find the first and the second most important variables)

Age, then Weight

5. (Total: 14 points)

This problem involves the Boston data set. We want to predict per capita crime rate using the other variables in this data set. In other words, per capita crime rate is the response, and the other variables are the predictors.

(a) (3 points) For each predictor, fit a simple linear regression model to predict the response. Describe your results. In which of the models is there a statistically significant association between the predictor and the response? Create some plots to back up your assertions.



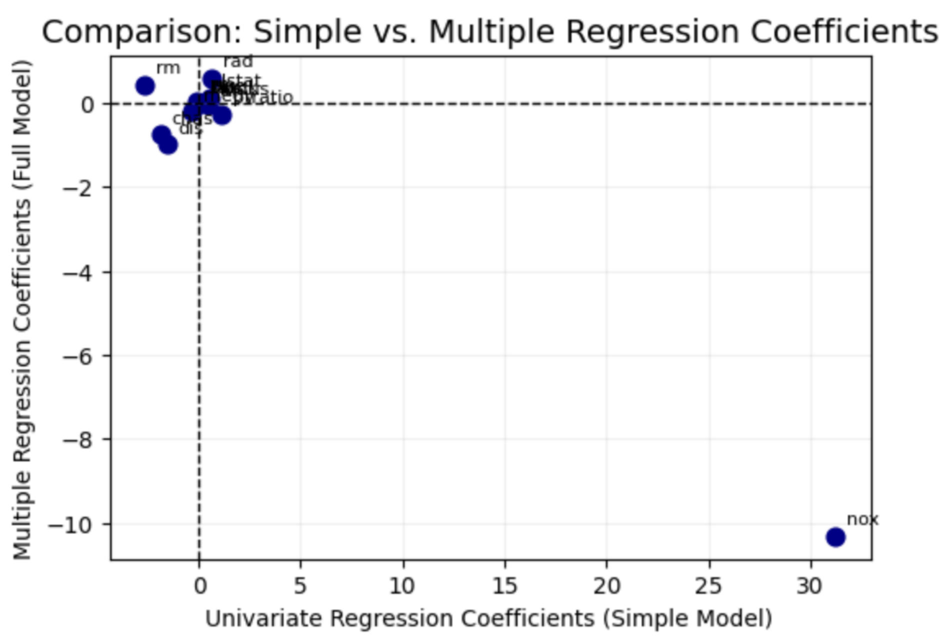
All predictors except for the chas has a model with significant association between the predictor and response. And the rad model fit the data most.

(b) (3 points) Fit a multiple regression model to predict the response using all of the predictors. Describe your results. For which predictors can we reject the null hypothesis $H_0 : \beta_j = 0$?

```
...                          OLS Regression Results
==============================================================================
Dep. Variable:                   crim   R-squared:                       0.454
Model:                            OLS   Adj. R-squared:                  0.440
Method:                 Least Squares   F-statistic:                     31.47
Date:                Sat, 21 Feb 2026   Prob (F-statistic):           1.57e-56
Time:                        02:45:03   Log-Likelihood:                -1653.3
No. Observations:                 506   AIC:                             3335.
Df Residuals:                     492   BIC:                             3394.
Df Model:                          13
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const         17.0332      7.235      2.354      0.019       2.818      31.248
zn             0.0449      0.019      2.394      0.017       0.008       0.082
indus         -0.0639      0.083     -0.766      0.444      -0.228       0.100
chas          -0.7491      1.180     -0.635      0.526      -3.068       1.570
nox          -10.3135      5.276     -1.955      0.051     -20.679       0.052
rm             0.4301      0.613      0.702      0.483      -0.774       1.634
age            0.0015      0.018      0.081      0.935      -0.034       0.037
dis           -0.9872      0.282     -3.503      0.001      -1.541      -0.433
rad            0.5882      0.088      6.680      0.000       0.415       0.761
tax           -0.0038      0.005     -0.733      0.464      -0.014       0.006
ptratio       -0.2711      0.186     -1.454      0.147      -0.637       0.095
black         -0.0075      0.004     -2.052      0.041      -0.015      -0.000
lstat          0.1262      0.076      1.667      0.096      -0.023       0.275
medv          -0.1989      0.061     -3.287      0.001      -0.318      -0.080
==============================================================================
Omnibus:                      666.613   Durbin-Watson:                   1.519
Prob(Omnibus):                  0.000   Jarque-Bera (JB):            84887.625
Skew:                           6.617   Prob(JB):                         0.00
Kurtosis:                      65.058   Cond. No.                     1.58e+04
==============================================================================
```

zn, dis, rad, black, medv

(c) (4 points) How do your results from (a) compare to your results from (b)? Create a plot displaying the univariate regression coefficients from (a) on the x-axis, and the multiple regression coefficients from (b) on the y-axis. That is, each predictor is displayed as a single point in the plot. Its coefficient in a simple linear regression model is shown on the x-axis, and its coefficient estimate in the multiple linear regression model is shown on the y-axis.

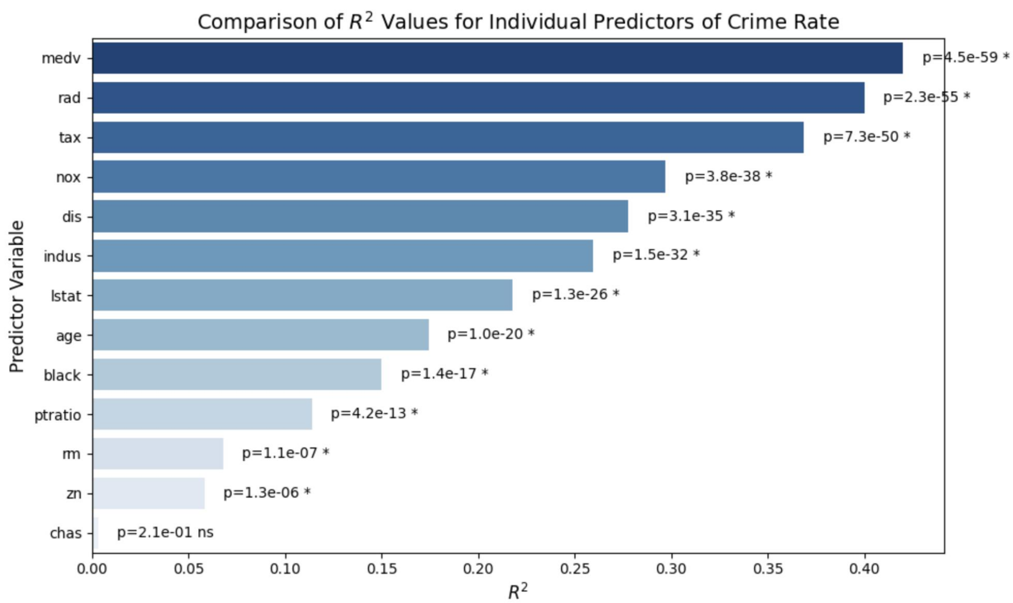


(d) (4 points) Is there evidence of non-linear association between any of the predictors and the response? To answer this question, for each predictor x, fit a model of the form

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \epsilon$$

```
models_5_non_linear = []
y_5 = df_5['crim']
for col in columns:
  X_5_non_linear = X_5[['const', col]].copy()
  X_5_non_linear[col + '2'] = X_5[col] ** 2
  X_5_non_linear[col + '3'] = X_5[col] ** 3

  model_5_non_linear = sm.OLS(y_5, X_5_non_linear).fit()
  models_5_non_linear.append(model_5_non_linear)
  print(model_5_non_linear.summary())
```



Yes. All predictors expect for chas could have a non-linear association with the response.